

Equidistribution of Twin Primes Across Admissible Residue Classes:

A Computational Census from $n = 3$ through $n = 8$

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Abstract

We present a full computational census of twin prime distributions across all twin-admissible residue classes modulo P_n for $n = 3$ through $n = 8$, where $P_n = \prod_{i=1}^n p_i$ is the n th primorial. At $n = 8$ ($P_8 = 9,699,690$), we enumerate all 378,675 twin-admissible classes across the range $[P_8, P_8^2] \approx [9.7 \times 10^6, 9.4 \times 10^{13}]$ — a 94-trillion-integer parallelised segmented sieve in 95.56 core-hours. We find that twin primes distribute across admissible classes with a sub-Poisson coefficient of variation: CV = 0.15% at $n = 8$, below the Poisson baseline of 0.17%, with zero classes empty. The bias between expected and actual total twin-pair count is $< 0.001\%$ at $n = 8$. The convergence from $n = 3$ (CV 2.44%) through $n = 8$ (CV 0.15%) is strictly monotonic and extends the variance suppression observed in the joint class occupancy conjecture into the 10^{14} scale. This census establishes the empirical foundation for studying equidistribution phenomena in prime k -tuple residue classes.

MSC 2020: 11N05, 11N35, 11Y11

1 Introduction

The Hardy–Littlewood k -tuples conjecture [1] predicts that prime constellations distribute across their admissible residue classes proportionally to their singular series. For twin primes, this is governed by the twin prime singular series $\mathfrak{S}(2) \approx 0.66016$. The twin-admissible residue classes modulo P_n are those $a \in (\mathbb{Z}/P_n\mathbb{Z})^\times$ such that $a, a + 2$ are both coprime to all $p \leq p_n$. There are $A_n = \prod_{2 < p \leq p_n} (p - 2)$ such classes.

A central question is whether these twin-admissible classes are uniformly populated, or whether arithmetic structure creates preferred classes. The joint class occupancy conjecture asserts equidistribution across all admissible classes. This paper provides empirical validation through a full computational census from $n = 3$ ($P_3 = 30$, 3 classes) through $n = 8$ ($P_8 = 9,699,690$, 378,675 classes), cross-verified against established aggregate twin prime counts [5].

This paper is Paper 2 in a three-part research program (RAPF trilogy):

- Paper 1: The discrete information-geometric framework establishing the modulo-6 lattice structure and Fisher information metric for twin primes.
- **Paper 2 (this work):** The equidistribution census providing empirical validation across six primorial scales ($n = 3$ through $n = 8$).

- Paper 3: Extension to prime triplets on the modulo-30 lattice ($\mathcal{L}_{30}$), with the exact Fisher metric $I(\lambda) = 225/[\lambda^4 \sinh^2(15/\lambda)]$ and an equidistribution census through $n = 7$ (9.3×10^7 triplets across 17,920 classes), demonstrating that the CV/CV_{Poisson} trajectory toward Poisson mirrors the twin prime pattern.

Relationship to companion papers. The discrete Fisher information metric and lattice structure derived in Paper 1 supply the structural foundation for the equidistribution phenomenon studied here. Paper 3 extends this framework to prime triplets on the modulo-30 lattice ($\mathcal{L}_{30}$), demonstrating that the sub-Poisson variance suppression observed here for twins ($k = 2$) manifests for higher-order constellations ($k = 3$), with an analogous finite-scale-to-Poisson relaxation trajectory. The algebraic non-vanishing analysis in Paper 1 establishes admissibility at all scales, while the empirical results of this paper demonstrate that twin-admissible classes are uniformly occupied across the six primordial scales studied. Together, Papers 1 and 2 form a complete architecture: the discrete geometry defines the structure, and the census confirms the equidistribution.

2 Joint Class Occupancy Conjecture

The empirical data supports the following conjecture about uniform class occupancy for twin-admissible residues at all scales $n \geq 3$:

Conjecture 1 (Joint Class Occupancy Conjecture). *For every $n \geq 3$ and every twin-admissible residue class $a \in \mathcal{A}_n$ (where $\mathcal{A}_n$ is the set of A_n twin-admissible classes modulo P_n), the twin prime pair count $\pi_2(x; P_n, a) > 0$ for all $x \geq P_n^2$. In other words, no twin-admissible class is empty.*

Our full census at $n = 8$ confirms this conjecture empirically: with 378,675 classes across 94 trillion integers, zero classes were empty. The minimum occupancy across all classes was 336,256 twin pairs and the maximum was 340,861 pairs (a tight $\sim 1.3\%$ spread around the mean of 338,680). The joint occupancy across the entire range was 128,249,570,494 twin pairs.

3 Methodology

3.1 Segmented Sieve Architecture

The census was performed using a parallelised segmented sieve over $[P_n, P_n^2]$. For $n = 8$ ($P_8 = 9,699,690$, range $[P_8, P_8^2] \approx 9.4 \times 10^{13}$), we used ~ 115 million 1 MB segments distributed across 8 threads. Each segment sieves using all base primes from 3 up to $\sqrt{P_8^2} = P_8$, then identifies twin pairs within each twin-admissible residue class.

3.2 Classification and Binning

For each twin pair $(p, p + 2)$ found in $[P_n, P_n^2]$, we classify the pair by its residue $p \bmod P_n$. Only twin-admissible classes (a such that $a, a + 2$ are both coprime to p_i for all $i \leq n$) are tracked. The census counts the number of twin pairs in each class, producing a histogram of length A_n from which we compute:

- Total twin-pair count (sum over all classes)
- Mean pairs per class = Total/ A_n
- Coefficient of variation = $100 \times \sigma/\mu$

- Minimum and maximum occupancy
- Number of empty classes

4 Results

4.1 Census Table

Table 1: Equidistribution of primes across twin-admissible classes mod P_n over range $[P_n, P_n^2]$.

n	P_n	A_n	Total Pairs	Mean (μ)	Bias %	CV	CV _{rand}
3	30	3	30	10.0	30.21	16.33	31.62
4	210	15	621	41.4	5.22	7.68	15.54
5	2,310	135	34,196	253.3	0.01	4.68	6.28
6	30,030	1,485	3,120,435	2,101.3	0.05	1.69	2.18
7	510,510	22,275	541,042,776	24,289.2	< 0.001	0.53 [†]	0.64
8	9,699,690	378,675	128,249,570,494	338,680.1	<0.001	0.15	0.17

[†] At $n = 7$, the range of class counts was $[23,779, 24,781]$. $\text{CV}_{\text{rand}} = 100/\sqrt{\mu}$ represents the independent Poisson baseline. Bias is calculated against the cumulative Hardy–Littlewood prediction $2C_2 \text{li}_2(P_n^2)$.

At $n = 8$, we computed all 378,675 twin-admissible classes across the full range $[P_8, P_8^2] \approx [9.7 \times 10^6, 9.4 \times 10^{13}]$ — a 94-trillion-integer census performed by a parallelised segmented sieve in 95.56 core-hours. The bias is below 0.001% (error of 813,232 among 128 billion twin pairs), the coefficient of variation is 0.15% (below the Poisson baseline of 0.17%), and zero classes were empty. This extends the sub-Poisson variance suppression observed since $n = 3$ into the 10^{14} scale.

4.2 Interpretation of the CV Baseline and Multinomial Constraints

The independent Poisson baseline $\text{CV}_{\text{rand}} = 100/\sqrt{\mu}$ represents the expected coefficient of variation if twin pairs were distributed across classes as independent random events. At $n = 8$, $\mu \approx 338,680$, giving $\text{CV}_{\text{rand}} \approx 0.17\%$.

Because the total number of twin pairs in the interval is fixed, the baseline distribution of class counts is technically multinomial rather than independent Poisson. The multinomial variance is $\sigma^2 = \mu(1 - 1/A_n)$. However, because the number of admissible classes A_n grows so rapidly, this fixed-total constraint has a vanishingly small effect. At $n = 8$ ($A_8 = 378,675$), the multinomial constraint reduces the expected CV by a factor of $\sqrt{1 - 1/378,675} \approx 0.999998$.

Therefore, the multinomial constraint accounts for a negligible fraction of the variance reduction. The observed CV of 0.15% (a $\sim 15\%$ suppression below the baseline) represents genuine structural anti-clumping across the primordial lattice.

5 Theta Sieve and Variance Suppression

The Theta Sieve methodology provides a computational tracking framework for understanding why twin primes show sub-Poisson variance across admissible classes. The sieve identifies regions of arithmetic space where the twin pair density is suppressed relative to random expectation. At $n = 8$, the 0.15% CV represents a stable variance suppression of $\sim 15\%$ below the Poisson baseline of 0.17%, consistent with the trend observed from $n = 3$ through $n = 7$.

This sub-Poisson variance pattern suggests that twin primes are not merely distributed randomly across admissible classes [3], but exhibit a structural anti-clumping behavior that becomes more pronounced at larger primorial scales [2]. This suppression may reflect the same arithmetic rigidity captured by the modulo-6 lattice structure in Paper 1, but the direct analytical connection remains an open area of inquiry.

6 Algorithmic Implementation

6.1 Parallelization Architecture

The census is inherently parallelizable: each segment of integers is processed independently. The implementation distributes work across T threads, each processing a contiguous segment of the range $[P_n, P_n^2]$. For $n = 8$, we used 8 threads, each processing ~ 14.5 million segments. Each thread maintains a local counter for each twin-admissible residue class, and final aggregation combines the per-thread counts via reduction.

6.2 Segmented Processing

For ranges exceeding available memory, the interval is partitioned into segments of fixed size (typically 1 MB for $n = 8$). Each segment is sieved independently using all base primes up to $\sqrt{\text{upper}}$, with cross-segment boundaries handled by extending each segment by two positions to capture twin pairs that straddle the boundary. This mirrors the boundary correction methodology established in the discrete information-geometric framework of Paper 1.

7 Conclusion

The full census of twin-admissible residue classes from $n = 3$ through $n = 8$ confirms that twin primes distribute uniformly across admissible classes with a sub-Poisson coefficient of variation at all scales studied. At $n = 8$ ($P_8 = 9,699,690$), the census found all 378,675 classes occupied with CV 0.15%, below the Poisson baseline of 0.17%, across 128 billion twin pairs in a 94-trillion-integer range. Zero classes were empty, confirming the joint class occupancy conjecture empirically out to $X = 9.4 \times 10^{13}$.

Combined with the discrete information-geometric framework (Paper 1), which establishes the permanent statistical structure of the twin prime distribution on the modulo-6 lattice, this census, together with the triplet extension in Paper 3, provides the empirical foundation for studying equidistribution phenomena in prime k -tuples. The triplet census confirms that the tighter modular constraints of the modulo-30 lattice produce a qualitatively similar variance suppression pattern at small primorial scales, establishing the $\text{CV}/\text{CV}_{\text{Poisson}}$ trajectory that converges toward the Poisson limit for both $k = 2$ and $k = 3$ constellations.

Acknowledgments

Sieve implementations, computational results, and LaTeX source for this paper are available at <https://github.com/rnmbaker-spec/rapf-prime-framework>.

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References

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